AI_AC 20.3

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BATCH-41

Task 1 – Password Strength & Validation Check

PROMPT: Password Strength & Validation Check

Requirements:

1. Create a function to check the strength of a password based on the following criteria

- Minimum length of 8 characters.

- Contains at least one uppercase letter.

- Contains at least one lowercase letter.

- Contains at least one digit.

2. The function should return a message indicating whether the password is "Strong", "Moderate", or "Weak".

3. Handle edge cases (e.g., empty string, only special characters) appropriately

E: > ai ac.py > ...

```python
#Password Strength & Validation Check
# Requirements:
# 1. Create a function to check the strength of a password based on the following criteria
#    - Minimum length of 8 characters.
#    - Contains at least one uppercase letter.
#    - Contains at least one lowercase letter.
#    - Contains at least one digit.
# 2. The function should return a message indicating whether the password is "Strong", "Moderate", or "Weak".
# 3. Handle edge cases (e.g., empty string, only special characters) appropriately
def check_password_strength(password):
    """
    Check the strength of a given password based on specific criteria.
    Parameters:
    password (str): The password string to be evaluated.
    Returns:
    str: A message indicating whether the password is "Strong", "Moderate", or "Weak".
    Criteria:
    - Strong: Meets all criteria (length >= 8, contains uppercase, lowercase, and digit).
    - Moderate: Meets at least two of the criteria.
    - Weak: Meets fewer than two criteria.
    """
    # Initialize criteria counters
    has_uppercase = any(char.isupper() for char in password)
    has_lowercase = any(char.islower() for char in password)
    has_digit = any(char.isdigit() for char in password)
    # Check the length of the password
    is_long_enough = len(password) >= 8
```


```python
    """
    # Initialize criteria counters
    has_uppercase = any(char.isupper() for char in password)
    has_lowercase = any(char.islower() for char in password)
    has_digit = any(char.isdigit() for char in password)
    # Check the length of the password
    is_long_enough = len(password) >= 8
```


```python
    # Count how many criteria are met
    criteria_met = sum([has_uppercase, has_lowercase, has_digit, is_long_enough])
    # Determine password strength based on criteria met
    if criteria_met == 4:
        return "Strong"
    elif criteria_met >= 2:
        return "Moderate"
    else:
        return "Weak"
# Example usage:
if __name__ == "__main__":
    test_password = "yashwanth@2486"
    print(f"The password '{test_password}' is: {check_password_strength(test_password)}")
```

OUTPUT:

```
PS C:\Users\yashw\AppData\Local\Programs\Microsoft VS Code> & C:\Users\yashw\AppData
y"
The password 'Password@2486' is: Strong
PS C:\Users\yashw\AppData\Local\Programs\Microsoft VS Code>
```

Task 2 – Insecure File Upload Handling

PROMPT: # Requirements:

1. Create a function to handle file uploads securely.

2. Validate the file type (e.g., only allow .jpg, .png, .pdf).

3. Check the file size (e.g., limit to 5MB).

4. Sanitize the file name to prevent directory traversal attacks.

5. Store the file in a designated directory with a unique name to avoid overwriting

```python
#Insecure File Upload handling
# Requirements:
# 1. Create a function to handle file uploads securely.
# 2. Validate the file type (e.g., only allow .jpg, .png, .pdf).
# 3. Check the file size (e.g., limit to 5MB).
# 4. Sanitize the file name to prevent directory traversal attacks.
# 5. Store the file in a designated directory with a unique name to avoid overwriting
import os
import uuid
def secure_file_upload(file_name, file_content):
    """
    Handle file uploads securely by validating file type, size, and sanitizing the file name.
    Parameters:
    file_name (str): The original name of the uploaded file.
    file_content (bytes): The content of the uploaded file.
    Returns:
    str: A message indicating the result of the upload process.
    Criteria:
    - Allowed file types: .jpg, .png, .pdf
    - Maximum file size: 5MB
    - Sanitized file name to prevent directory traversal attacks.
    - Unique file name to avoid overwriting existing files.
    """
    # Define allowed file extensions and maximum file size
    allowed_extensions = {'.jpg', '.png', '.pdf'}
    max_file_size = 5 * 1024 * 1024  # 5MB in bytes
    # Extract the file extension
    _, file_extension = os.path.splitext(file_name)
    # Validate the file type
    if file_extension.lower() not in allowed_extensions:
        return "Invalid file type. Only .jpg, .png, and .pdf are allowed."
```

```python
    if len(file_content) > max_file_size:
        return "File size exceeds the limit of 5MB."
    # Sanitize the file name to prevent directory traversal attacks
    sanitized_file_name = os.path.basename(file_name)
    # Generate a unique file name to avoid overwriting existing files
    unique_file_name = f"{uuid.uuid4()}{file_extension}"
    # Define the directory to store uploaded files
    upload_directory = "uploads"
    # Ensure the upload directory exists
    os.makedirs(upload_directory, exist_ok=True)
    # Save the file securely
    with open(os.path.join(upload_directory, unique_file_name), 'wb') as f:
        f.write(file_content)
    return f"File uploaded successfully as {unique_file_name}."
# Example usage:
if __name__ == "__main__":
    # Simulate file upload
    test_file_name = "example.pdf"
    test_file_content = b"%PDF-1.4 example content"  # Simulated file content in bytes
    print(secure_file_upload(test_file_name, test_file_content))
```

OUTPUT:

```
PS C:\Users\yashw\AppData\Local\Programs\Microsoft VS Code> & C:\Users\yashw\AppData\Local\Programs\Python\Python313\python.exe "e:/ai ac.p
y"
File uploaded successfully as 8f6d3cc4-34ce-460c-a721-87a0e9f82f01.pdf.
PS C:\Users\yashw\AppData\Local\Programs\Microsoft VS Code>
```

Task 3 – Real-Time Application: API Security Review

Requirements:

1. Create a function to review API security based on common vulnerabilities.

2. Check for vulnerabilities such as:

- SQL Injection

- Cross-Site Scripting (XSS)

- Insecure Direct Object References (IDOR)

3. The function should return a report indicating the presence of any vulnerabilities and recommendations for mitigation

```python

#Real-Time Application: API Security review
# Requirements:
# 1. Create a function to review API security based on common vulnerabilities.
# 2. Check for vulnerabilities such as:
#    - SQL Injection
#    - Cross-Site Scripting (XSS)
#    - Insecure Direct Object References (IDOR)
# 3. The function should return a report indicating the presence of any vulnerabilities and recommendations for mitigation.
def review_api_security(api_endpoint):
    """
    Review API security based on common vulnerabilities such as SQL Injection, XSS, and IDOR.
    Parameters:
    api_endpoint (str): The API endpoint to be reviewed for security vulnerabilities.
    Returns:
    str: A report indicating the presence of any vulnerabilities and recommendations for mitigation.
    """
    # Simulated vulnerability checks (In a real implementation, this would involve more complex logic)
    vulnerabilities = []
    # Check for SQL Injection vulnerability
    if "SELECT" in api_endpoint.upper() or "DROP" in api_endpoint.upper():
        vulnerabilities.append("SQL Injection detected. Recommendation: Use parameterized queries.")
    # Check for Cross-Site Scripting (XSS) vulnerability
    if "<script>" in api_endpoint.lower():
        vulnerabilities.append("XSS detected. Recommendation: Sanitize user input and use Content Security Policy (CSP).")
    # Check for Insecure Direct Object References (IDOR) vulnerability
    if "/user/" in api_endpoint.lower() and any(char.isdigit() for char in api_endpoint):
        vulnerabilities.append("IDOR detected. Recommendation: Implement proper access controls and authorization checks.")
    # Generate the report
    if vulnerabilities:
        report = "Vulnerabilities found:\n" + "\n".join(vulnerabilities)
    else:
        report = "No vulnerabilities detected. The API endpoint appears to be secure."
    return report
```

```python
# Example usage:
if __name__ == "__main__":
    test_api_endpoint = "/user/123?name=<script>alert('XSS')</script>"
    print(review_api_security(test_api_endpoint))
```

OUTPUT:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   powershell

PS C:\Users\yashw\AppData\Local\Programs\Microsoft VS Code> & C:\Users\yashw\AppData\Local\Programs\Python\Python313\python.exe "e:/ai ac.py"
Vulnerabilities found:
XSS detected. Recommendation: Sanitize user input and use Content Security Policy (CSP).
IDOR detected. Recommendation: Implement proper access controls and authorization checks.
PS C:\Users\yashw\AppData\Local\Programs\Microsoft VS Code>
```

Task 4 – Cross-Site Scripting (XSS) Check

PROMPT: Requirements:

1. Create a function to check for potential XSS vulnerabilities in user input.

2. The function should analyze the input for common XSS attack vectors (e.g., <script> tags, event handlers).

3. The function should return a message indicating whether the input is safe or potentially vulnerable to XSS attacks

```python
    print(review_api_security(test_api_endpoint))'''
#Cross-Site Scripting (XSS) Check
# Requirements:
# 1. Create a function to check for potential XSS vulnerabilities in user input.
# 2. The function should analyze the input for common XSS attack vectors (e.g., <script> tags, event handlers).
# 3. The function should return a message indicating whether the input is safe or potentially vulnerable to XSS attacks.
def check_xss_vulnerability(user_input):
    """
    Check for potential Cross-Site Scripting (XSS) vulnerabilities in user input.
    Parameters:
    user_input (str): The input string to be analyzed for XSS vulnerabilities.
    Returns:
    str: A message indicating whether the input is safe or potentially vulnerable to XSS attacks.
    """
    # Define common XSS attack vectors
    xss_attack_vectors = ["<script>", "onerror=", "onload=", "javascript:"]
    # Check if any of the attack vectors are present in the user input
    for vector in xss_attack_vectors:
        if vector in user_input.lower():
            return "Potential XSS vulnerability detected. Input is potentially unsafe."
    return "Input is safe from XSS vulnerabilities."
# Example usage:
if __name__ == "__main__":
    test_input = "<script>alert('XSS')</script>"
    print(check_xss_vulnerability(test_input))
```

OUTPUT:

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   powershell
                                   Open file in editor (ctrl + click)

PS C:\Users\yashw\AppData\Local\Programs\Microsoft VS Code> & C:\Users\yashw\AppData\Local\Programs\Python\Python313\python.exe "e:/ai ac.py"
Potential XSS vulnerability detected. Input is potentially unsafe.
PS C:\Users\yashw\AppData\Local\Programs\Microsoft VS Code>
```

Task 5 – SQL Injection Prevention

PROMPT: Requirements:

1. Create a function to prevent SQL injection attacks by sanitizing user input.

2. The function should escape special characters and use parameterized queries

```python

# SQL Injection Prevention
# Requirements:
# 1. Create a function to prevent SQL injection attacks by sanitizing user input.
# 2. The function should escape special characters and use parameterized queries.
def prevent_sql_injection(user_input):
    """
    Prevent SQL injection attacks by sanitizing user input.
    Parameters:
    user_input (str): The input string to be sanitized for SQL injection prevention.
    Returns:
    str: A sanitized version of the input string that is safe for use in SQL queries.
    """
    # Escape special characters commonly used in SQL injection attacks
    sanitized_input = user_input.replace("'", "''").replace(";", "").replace("--", "")
    return sanitized_input
# Example usage:
if __name__ == "__main__":
    test_input = "O'Reilly; DROP TABLE users; --"
    print(prevent_sql_injection(test_input))
```

OUTPUT:

```python
print(prevent_sql_injection(test_input))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell

```
PS C:\Users\yashw\AppData\Local\Programs\Microsoft VS Code> & C:\Users\yashw\AppData\Local\Programs\Python\Python313\python.exe "e:/ai ac.py"
O''Reilly DROP TABLE users
PS C:\Users\yashw\AppData\Local\Programs\Microsoft VS Code>
```